

Outliers impact data analysis process badly. The result obtained differs from the actual result due to presence of outliers. Outliers affect the MEAN value of dataset.

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REMOVING OUTLIERS is crucial when it comes to data analysis or statistical analysis:-

1. Interquartile Range :-

$$IQR = Q_3 - Q_1 \quad \text{where } Q_3, \text{ Upper Whisker} \\ Q_1, \text{ Lower Whisker}$$

$$\therefore \begin{cases} \text{Data Point} > Q_3 + 1.5 \times IQR = \\ \text{Data Point} < Q_1 - 1.5 \times IQR = \end{cases} \text{OUTLIER}$$

2. Z-SCORE CALCULATION is quite useful for identifying outliers in a dataset. It calculates how many

$$Z = \left(\frac{X - \mu}{\sigma} \right)$$

SDs a point is from the mean.

• Typically, a Z-score beyond +3 or -3 is considered an outlier.

• हम हर datapoint के लिए Z-score calculate करेंगे ताकि जान सकें वो outlier हैं या नहीं।

